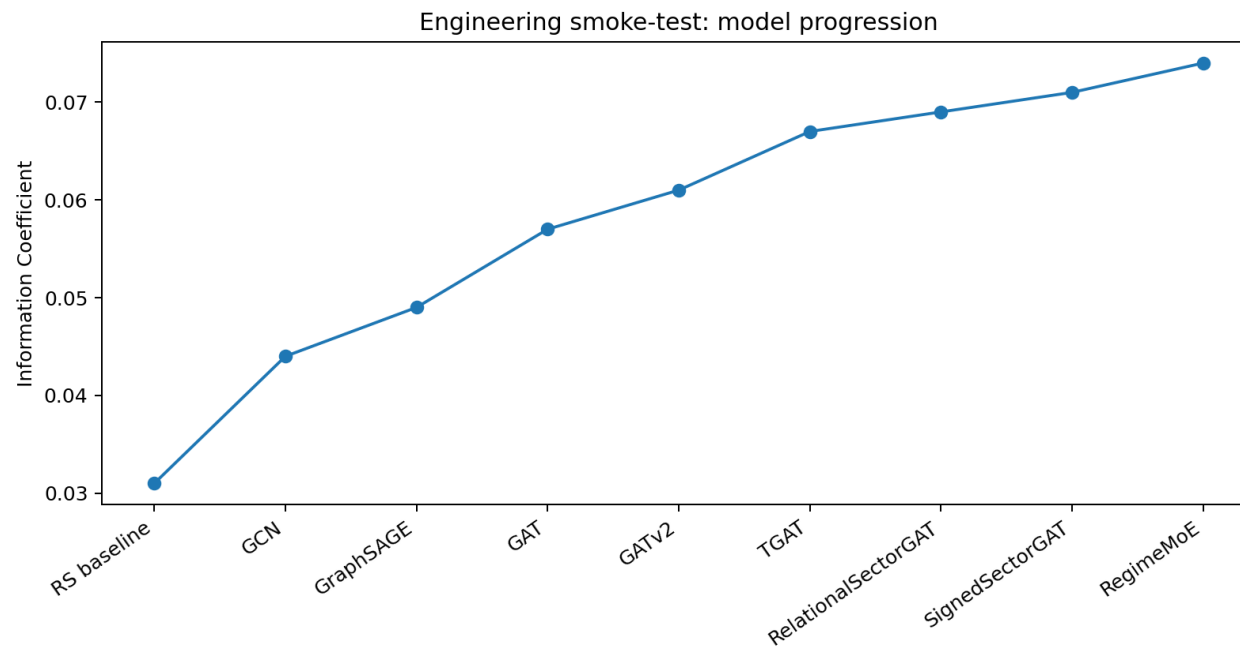


01 Temporal Graph Sector Rotation Network

Rank NSE sectoral indices by relative-strength gain/loss conditional on regime. Core principle: direction over price; explanation over black-box ranking.



02 The Cross-Sectional Question

Given the prevailing regime, identify sectors gaining relative strength and sectors losing it.
Output = ranked sectors + edge-level rationale + uncertainty state.

03 Why Static Correlation Fails

Static correlation is symmetric, averaged across regimes and dense. The target system must adapt to structural breaks, asymmetric lead-lag and sparse informative relationships.

04 Data Universe & Feature Stack

14 intended sectors + Nifty 50/500 + USDINR + 10Y Gilt + Brent. Features: 5/21/63/252D momentum, RS, volatility, beta, breadth, valuation, macro sensitivity.

05 Graph Construction

Hybrid prior graph + learned attention. Candidate edges: Pearson, partial correlation, mutual information, Granger lead-lag and flow/co-exposure.

06 Architecture Progression

GCN → GraphSAGE → GAT → GATv2 → TGAT → EvolveGCN → Relational/Signed → RegimeMoE. Complexity survives only if ablation supports it.

07 GAT / GATv2

Attention assigns different weights to different neighbours. Multi-head attention creates several relationship views; GATv2 provides more flexible query-dependent attention.

08 TGAT + Time Encoding

Learned time encoding is injected into node representations so attention depends on state and recency. Compare adaptation speed around structural breaks.

09 Dynamic / Heterogeneous / Signed Graphs

Relational edges: correlation / lead-lag / flows. Signed edges: positive / negative channels.

Dynamic edges: top-k similarity from current embeddings.

10 Uncertainty & Conformal Decision Rule

Act only when sector is top-k by mean score, inside conformal set, and below max predictive variance. Otherwise retain benchmark weight.

11 Edge-Level Explainability

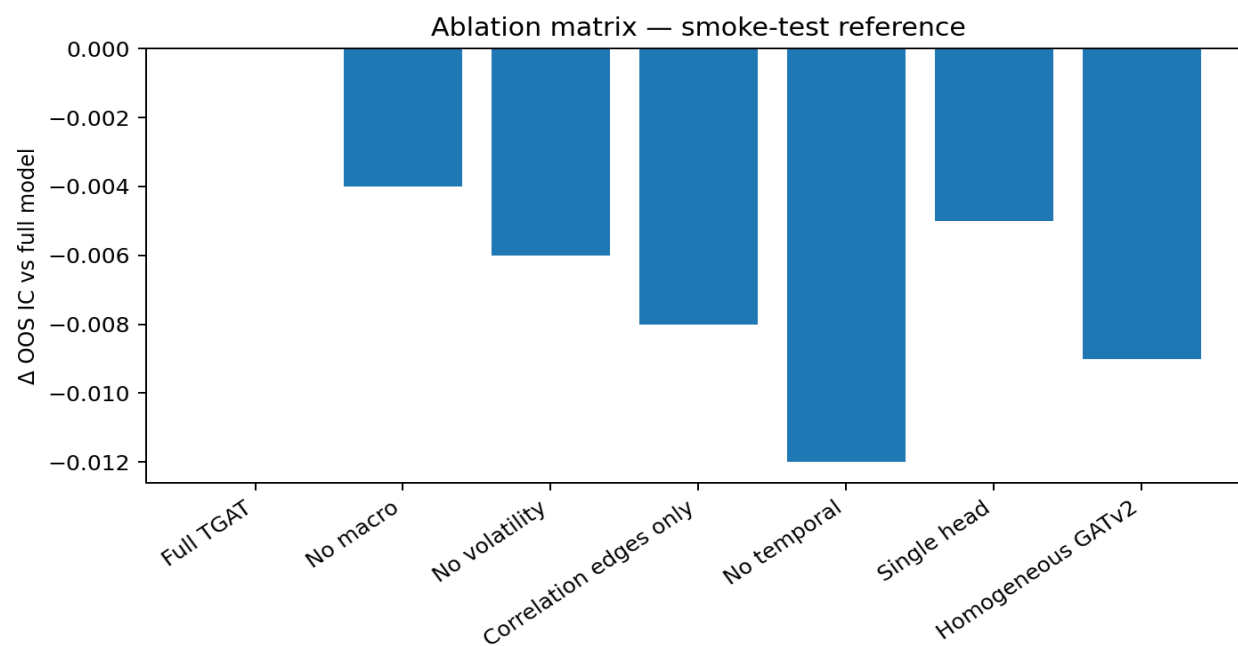
Three primitives: attention extraction, GNNExplainer, counterfactual edge ablation. Stability: Kendall tau, top-k persistence, attention entropy.

12 Validation & Ranking Metrics

Primary: cross-sectional IC. Secondary: Rank IC, NDCG@k, top-k hit rate, ICIR. All reported under expanding-window time-series CV.

13 Ablation Programme

Remove macro, volatility, edge types, temporal encoding, extra heads and heterogeneous structure one at a time. Re-measure OOS IC on identical folds.



14 Adversarial & Structural Stress Tests

Noise perturbation + edge perturbation. Structural tests: IL&FS; decorrelation, OOD regime, liquidity-shock isolation, contagion-path tracing.

15 Case Study Framework

IL&FS; 2018, 2020 IT rerating, 2023–24 PSU Bank cluster, 2024 election, 2013 Taper Tantrum. Test attention reassignment, not only returns.

16 Overlay Backtest & Costs

Top-k / linear / concentrated overlays. Report after-cost CAGR, active return, tracking error, IR, drawdown, turnover, sector and regime attribution.

17 Sector Compass Simulation

Eight levels from RS Ranker to Regulator's Choice; stress-test mode replays historical rotations; coach mode reviews explanations.

18 Audit-Ready Handover

Six mandatory deliverables plus 10-minute video. Model registry stores data hash, configuration, code commit, seed, attention lineage and post-trade attribution.